Project Title: Inventory & Warehouse Management System (IWMS)

Document Title: Architecture and Components Design Document

Version: 1.0

Date: 22|01|26

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Brief Project Overview:

The Inventory & Warehouse Management System (IWMS) is designed to manage stock inflow and outflow, track inventory, generate reports, and support warehouse operations efficiently.

Version History

Version	Date	Author	Changes Made
1.0	22|01|26	Deepanshu Gupta	Initial Draft

1. Section :

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Section 1 – Context & Requirements Integration

1.1 Introduction and Purpose:

This document describes the architecture and component design of the Inventory & Warehouse Management System (IWMS). It explains how the system is structured, how components interact, and how the design satisfies the functional and non-functional requirements identified during the requirements analysis phase.

1.2 Project Overview:

The Inventory & Warehouse Management System (IWMS) is designed to automate warehouse operations including stock management, supplier tracking, batch monitoring, and inventory reporting.

The system targets warehouse administrators, staff members, and managers who need a reliable and scalable system to monitor stock levels, prevent shortages, and generate valuation reports.

The primary objective of the system is to improve operational efficiency, reduce manual errors, and provide real-time inventory insights.

1.3 Key Requirements Summary

Functional Requirements:

- Stock In and Stock Out management
- Low stock alert system
- Batch & expiry tracking
- Supplier management
- Inventory valuation reports

Non-Functional Requirements:

- High performance for bulk transactions
- Scalability for large warehouses
- Data accuracy and validation
- 99.5% system availability
- Maintainability through modular design

1.4 Assumptions and Constraints

Assumptions:

- System users have basic computer knowledge
- Reliable internet connection is available
- Database server is accessible

Constraints:

- Must use relational database (MySQL/PostgreSQL)
- Limited budget and hardware resources
- Must comply with basic data security policies

1.5 Requirements Diagrams

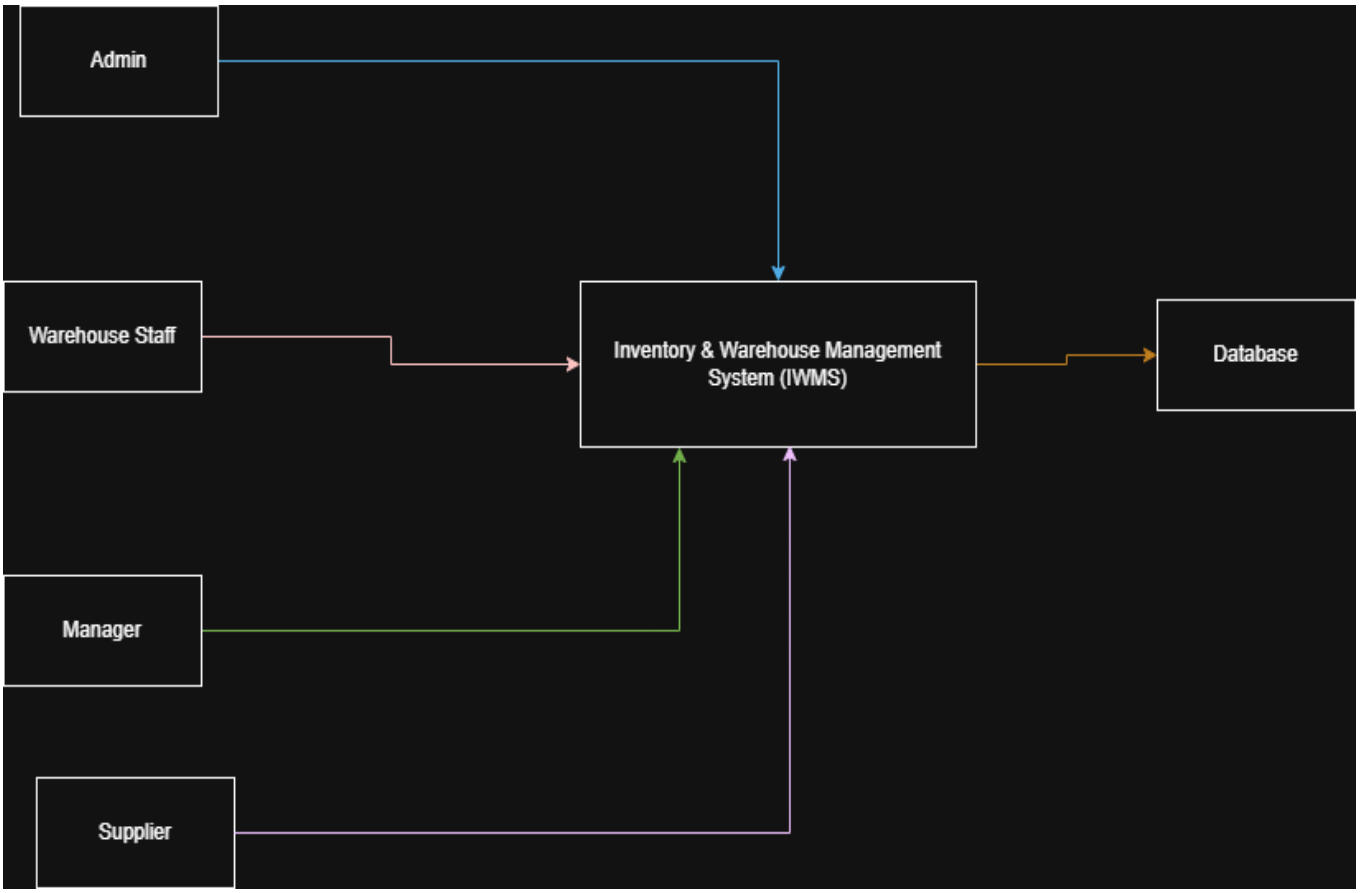


Figure 1-1: System Context Diagram

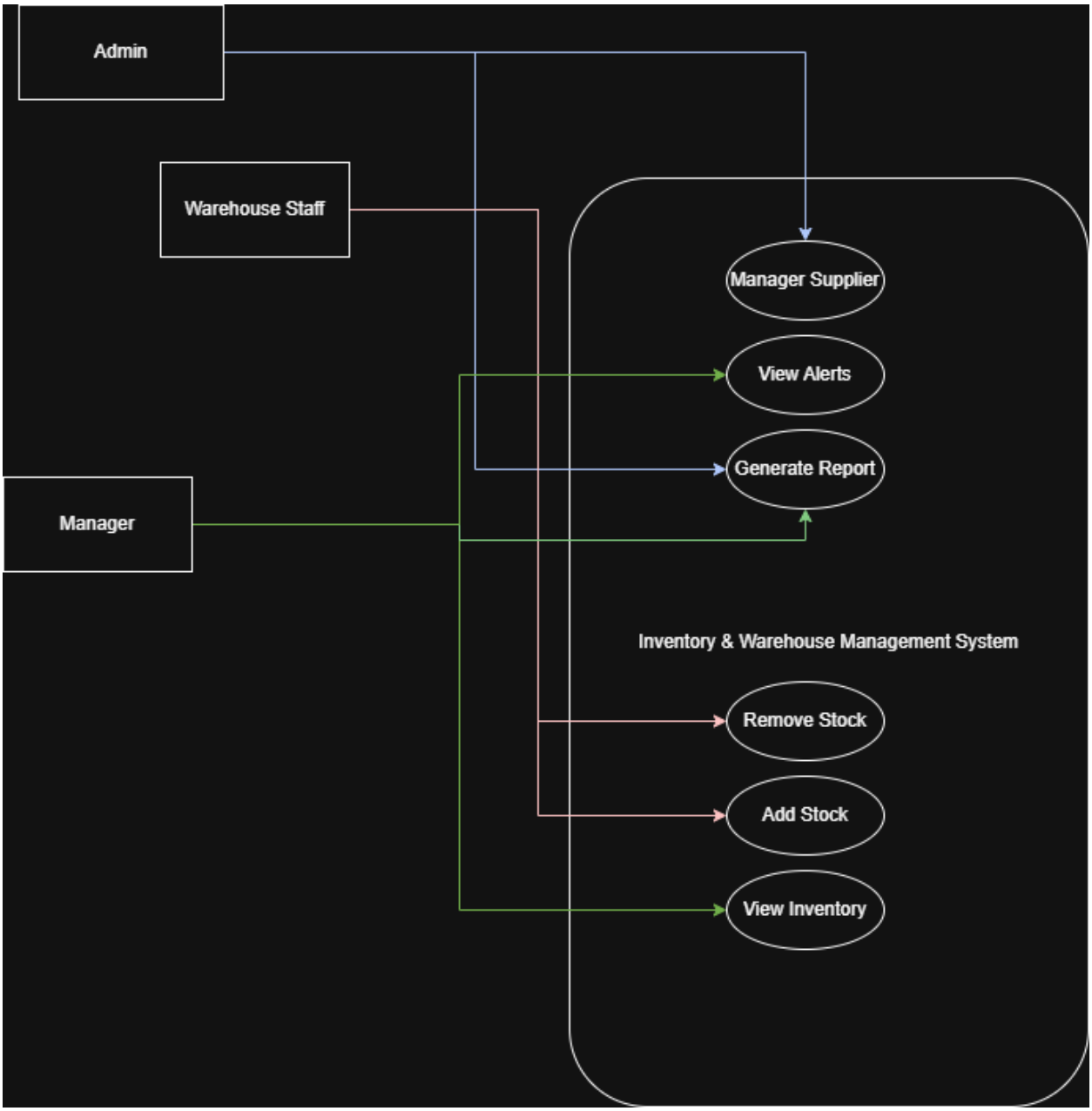


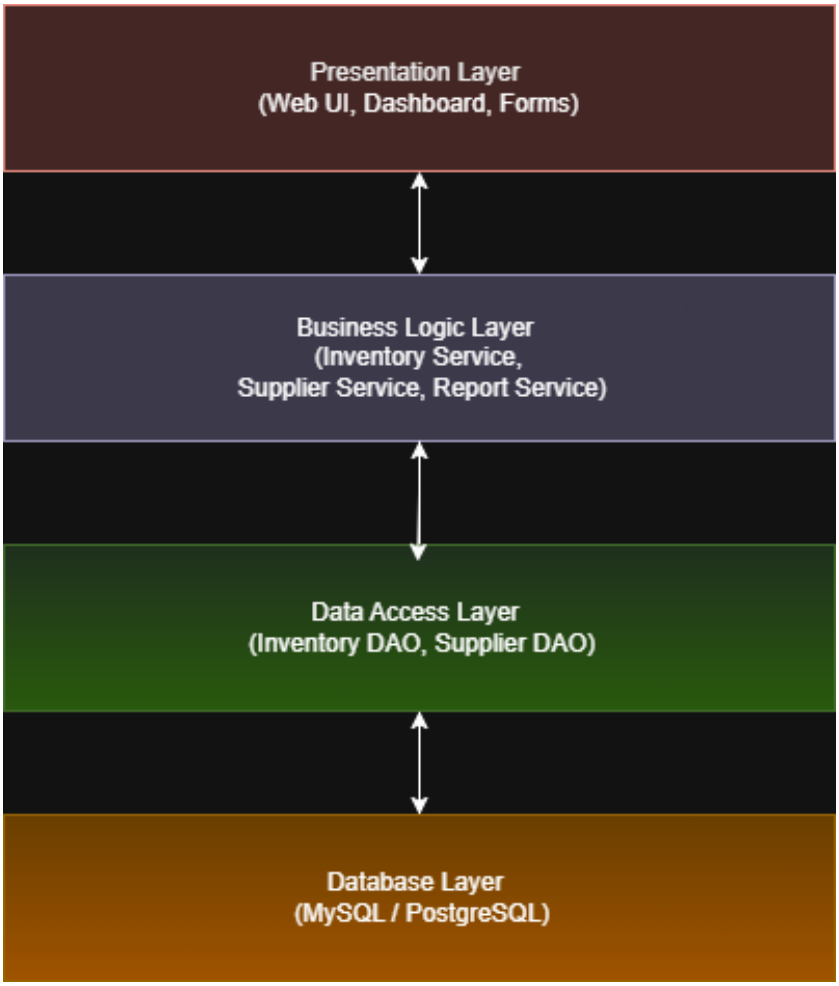
Figure 1-2: Use Case Diagram

Section 2 – System Architecture Design

2.1 High-Level Design

The system follows a layered architecture model consisting of Presentation Layer, Business Logic Layer, Data Access Layer, and Database Layer. This separation ensures modularity, scalability, and maintainability.

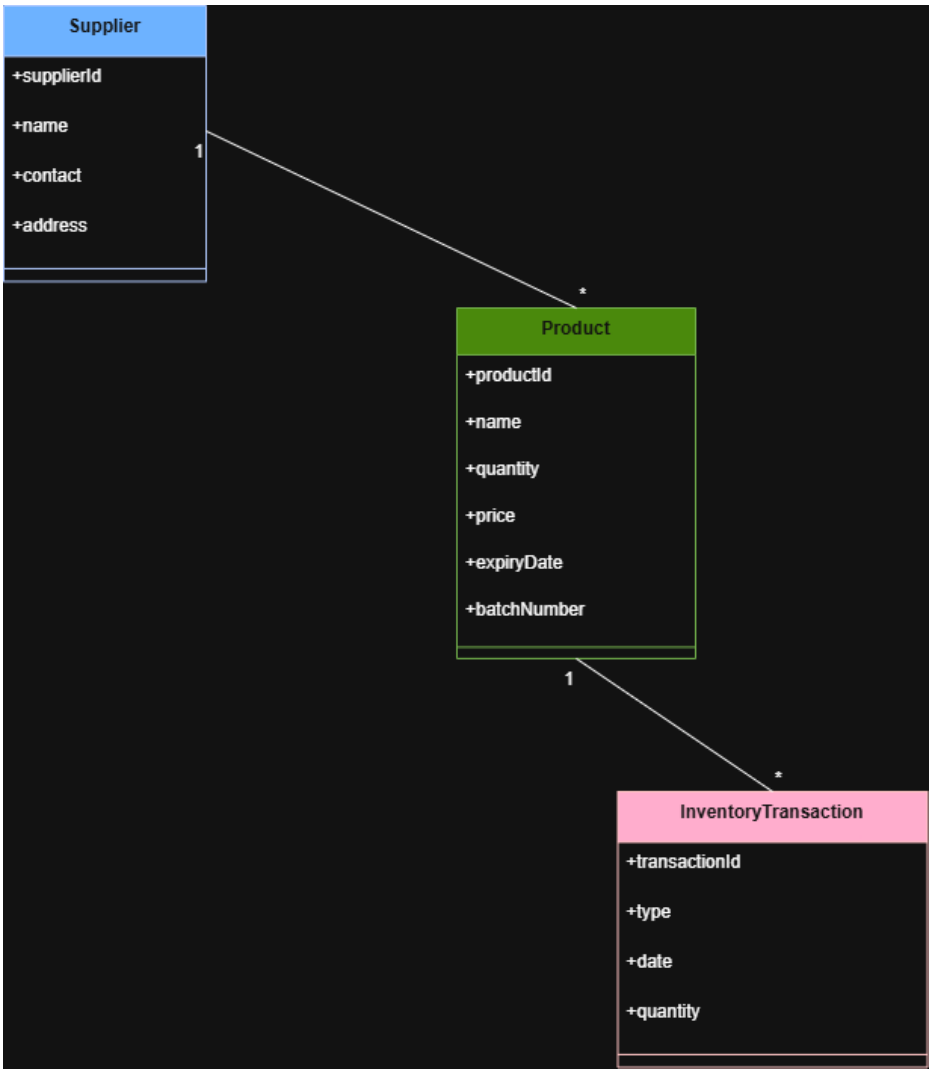
2.1.1 Layered Architecture View



The layered architecture separates user interface, business rules, and data storage responsibilities to improve system flexibility.

2.2 Component-Level Design – Inventory Module

The Inventory Module is the core component of the system responsible for managing stock inflow and outflow operations. It handles inventory updates, transaction logging, and threshold validation for low-stock alerts. The module follows a layered structure consisting of Controller, Service, and Data Access components. It interacts directly with the central database and communicates with the Alert Module for monitoring stock levels.



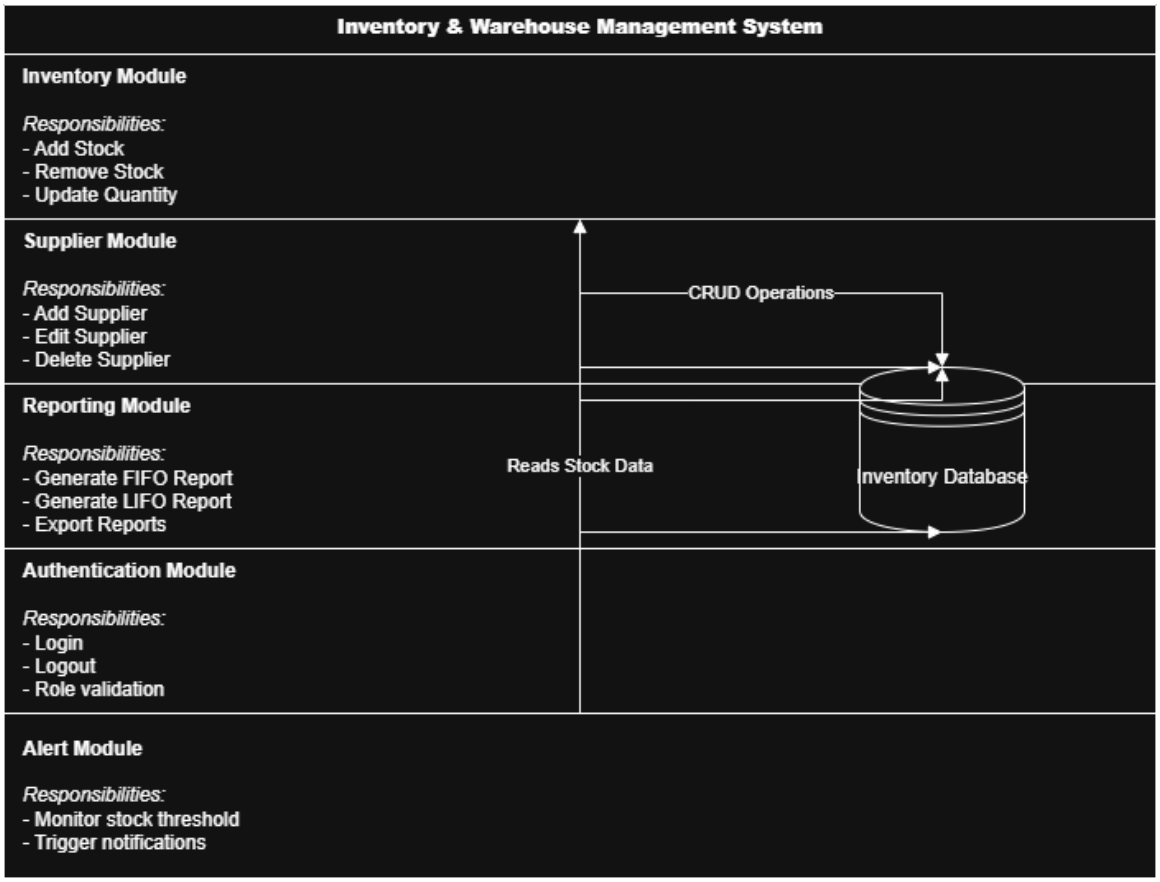


Figure 2-2: Component Diagram

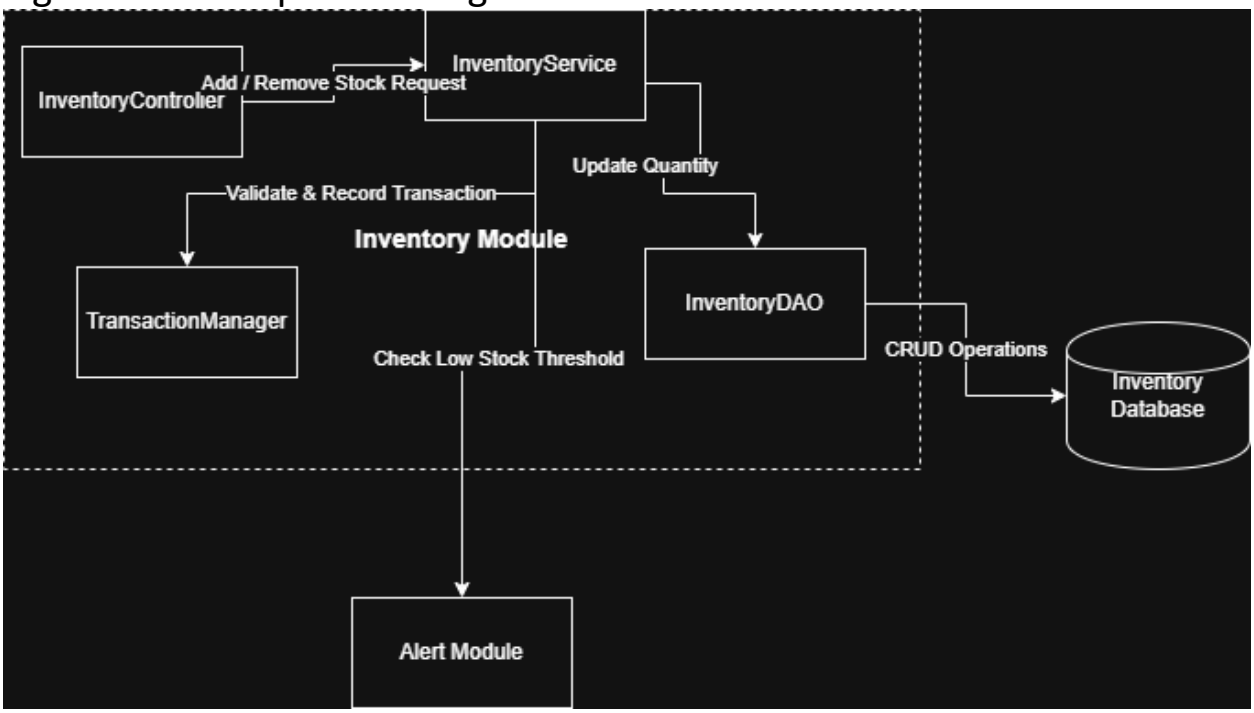


Figure 2-3: Component Diagram-Inventory Module

2.3 Component-Level Design – Supplier Module

The Supplier Module handles supplier registration, updating supplier information, and linking suppliers to stock records

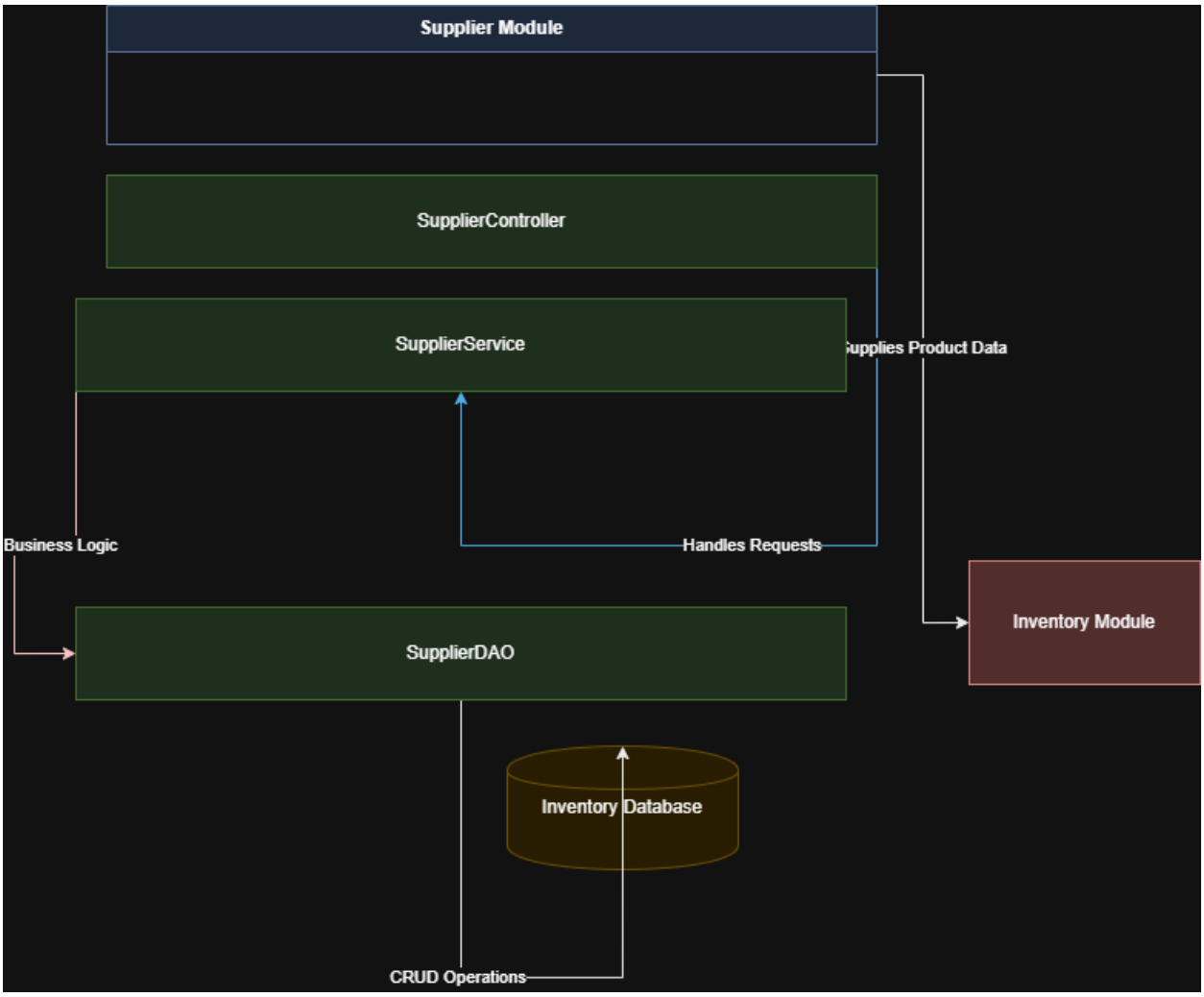


Figure 2-4: Component Diagram – Supplier Module

2.4 Component-Level Design – Reporting Module

The Reporting Module is responsible for generating inventory valuation reports using FIFO, LIFO, and Weighted Average methods. It retrieves inventory data from the database, applies valuation algorithms, and presents formatted reports for users. The module also supports exporting reports in PDF and Excel formats.

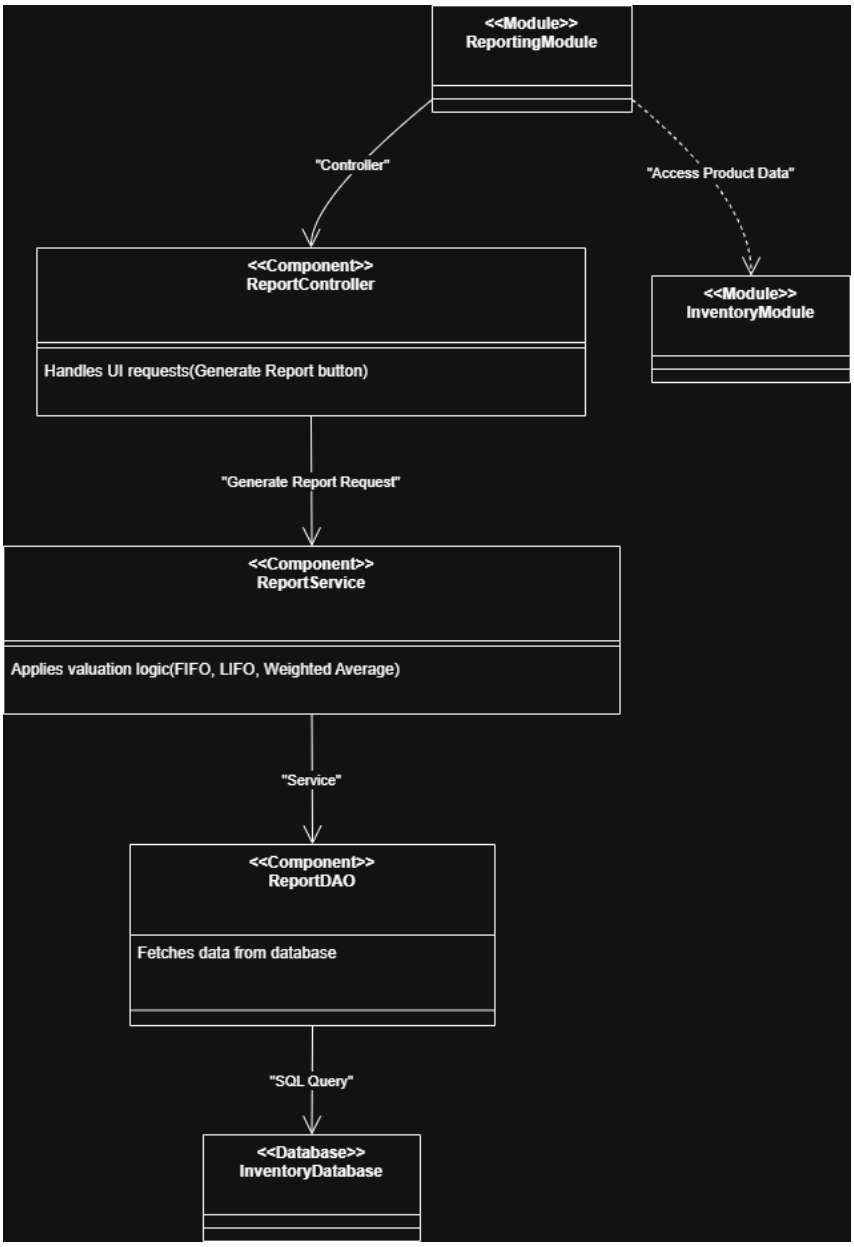


Figure 2-5: Component Diagram – Reporting Module

2.5 Behavioral Design – Sequence Diagram

The sequence diagram illustrates the interaction between Warehouse Staff, the IWMS system, and the database during the stock addition process. The system validates input data, updates inventory records, and confirms successful transaction to the user.

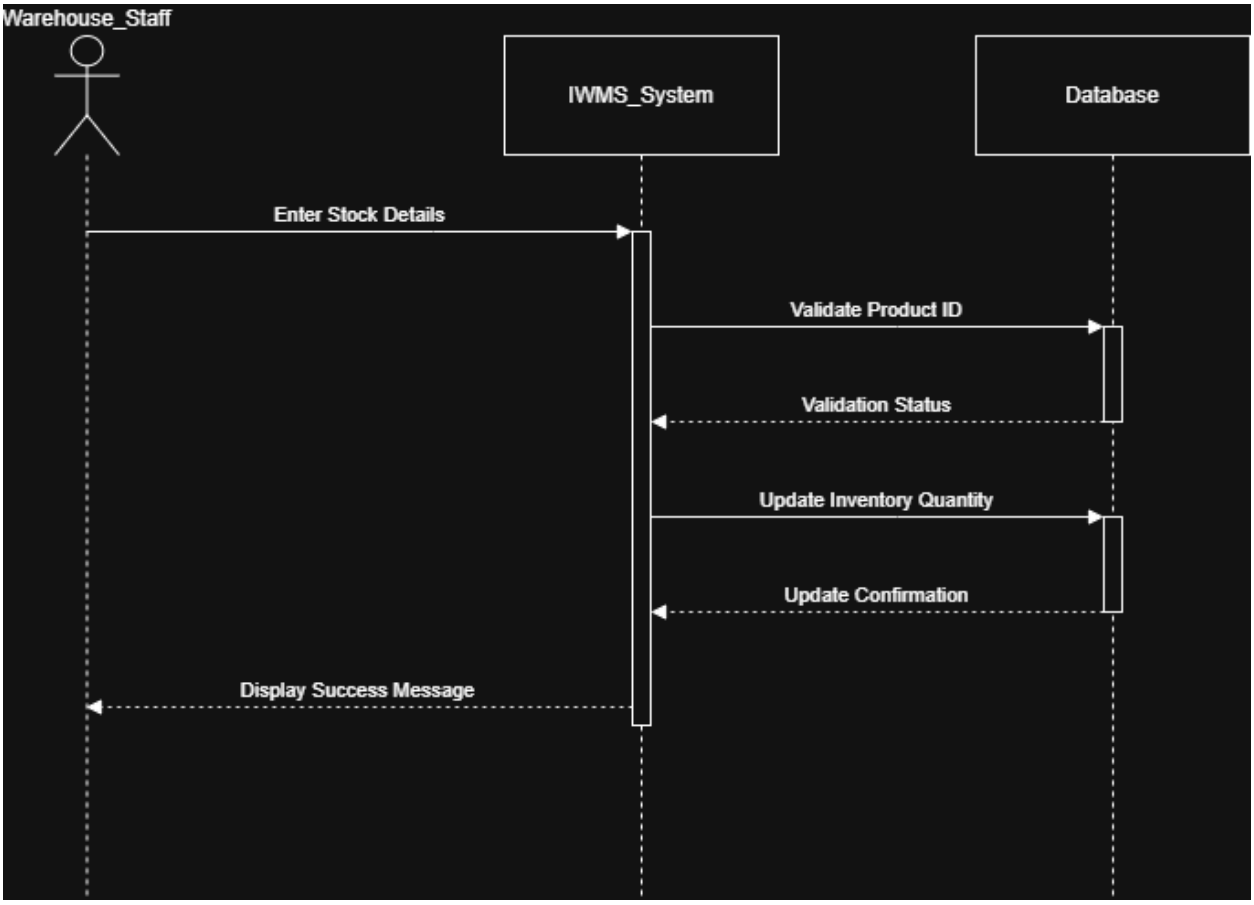
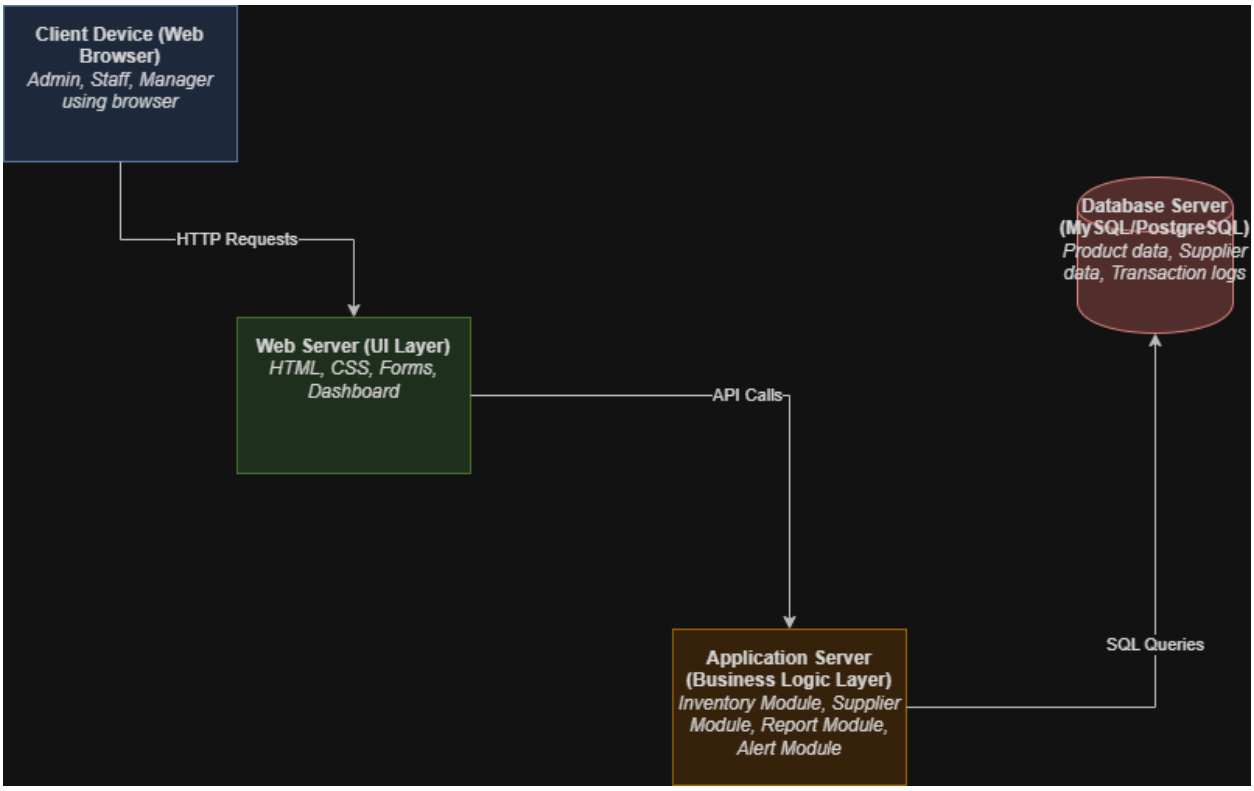


Figure 2-6: Sequence Diagram – Add Stock Process

2.6 Deployment Diagram

The deployment diagram represents the physical architecture of the IWMS system. Users access the system through a web browser, which communicates with the web server. The application server processes business logic and interacts with the database server for data storage and retrieval.



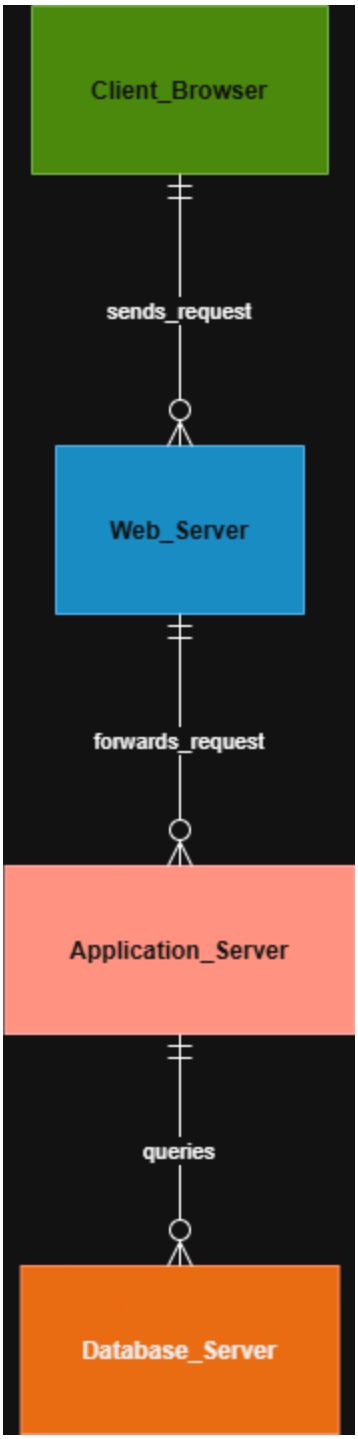


Figure 2-7: Deployment Diagram – IWMS

2.7 Design Evolution / Refactoring Notes

Initially, the system was designed as a single monolithic module. During refinement, the architecture was modularized into separate components (Inventory, Supplier, Reporting) to improve maintainability, scalability, and future extensibility.

Section 3 – UI/UX Design

3.1 GUI Interaction Flow

The GUI interaction flow represents how users navigate through the system to perform operations such as login, managing stock, generating reports, and viewing alerts.

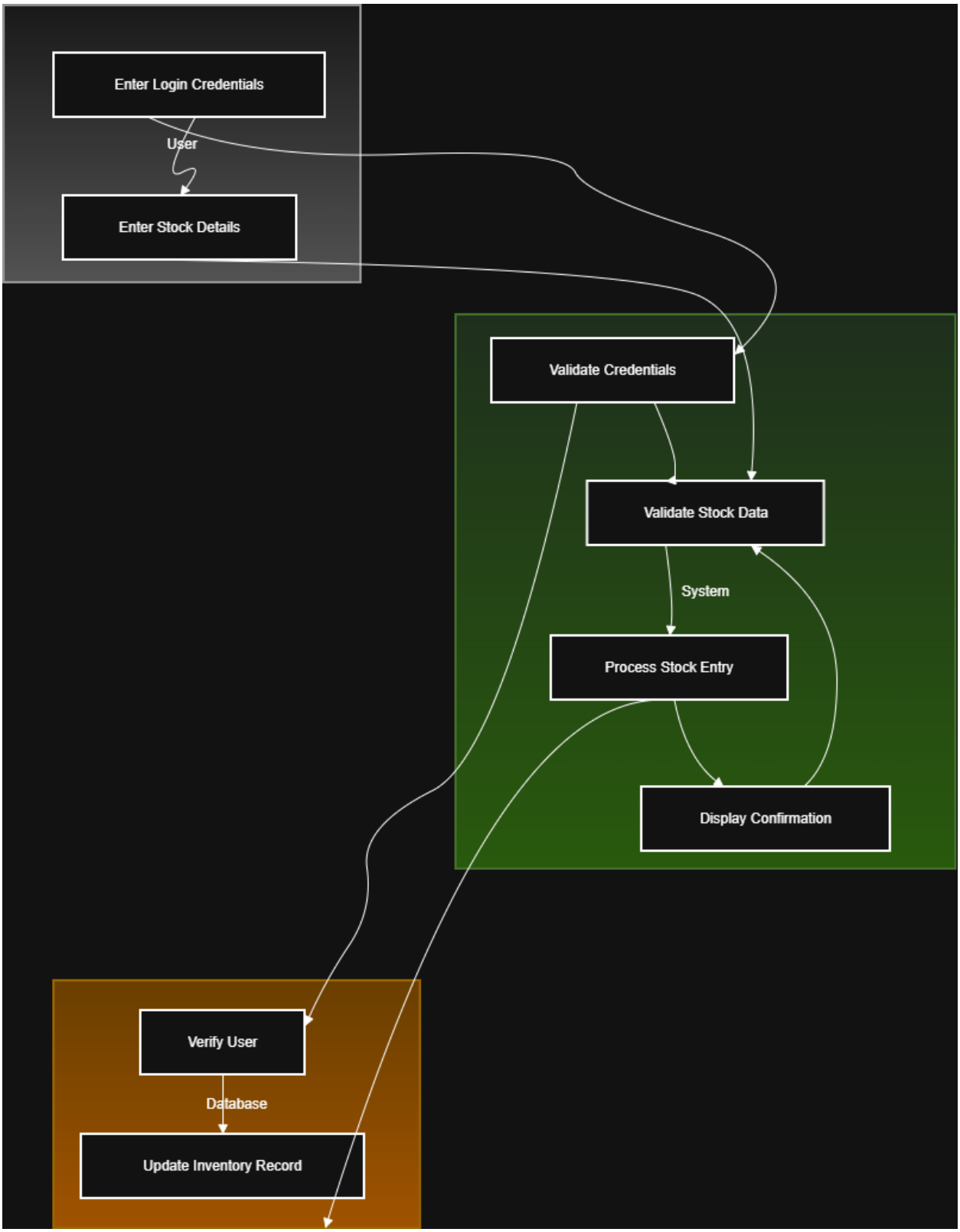


Figure 3-1: GUI Interaction Flow (Swimlane Diagram)

3.2 GUI Layout Prototypes

The user interface prototypes were manually designed to represent key workflows and screen layouts.

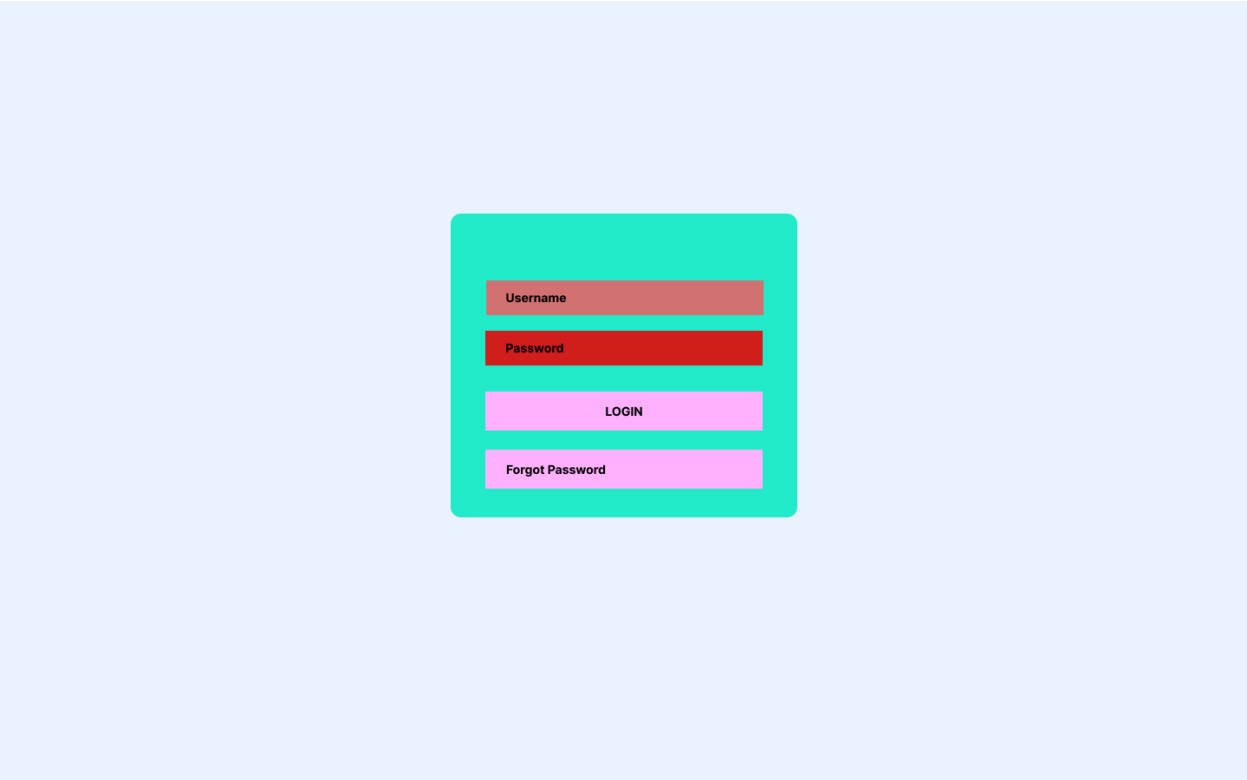


Figure 3-2: Login Screen

Figure 3-3: Dashboard Screen

Figure 3-4: Stock Entry Screen

Figure 3-5: Report Screen

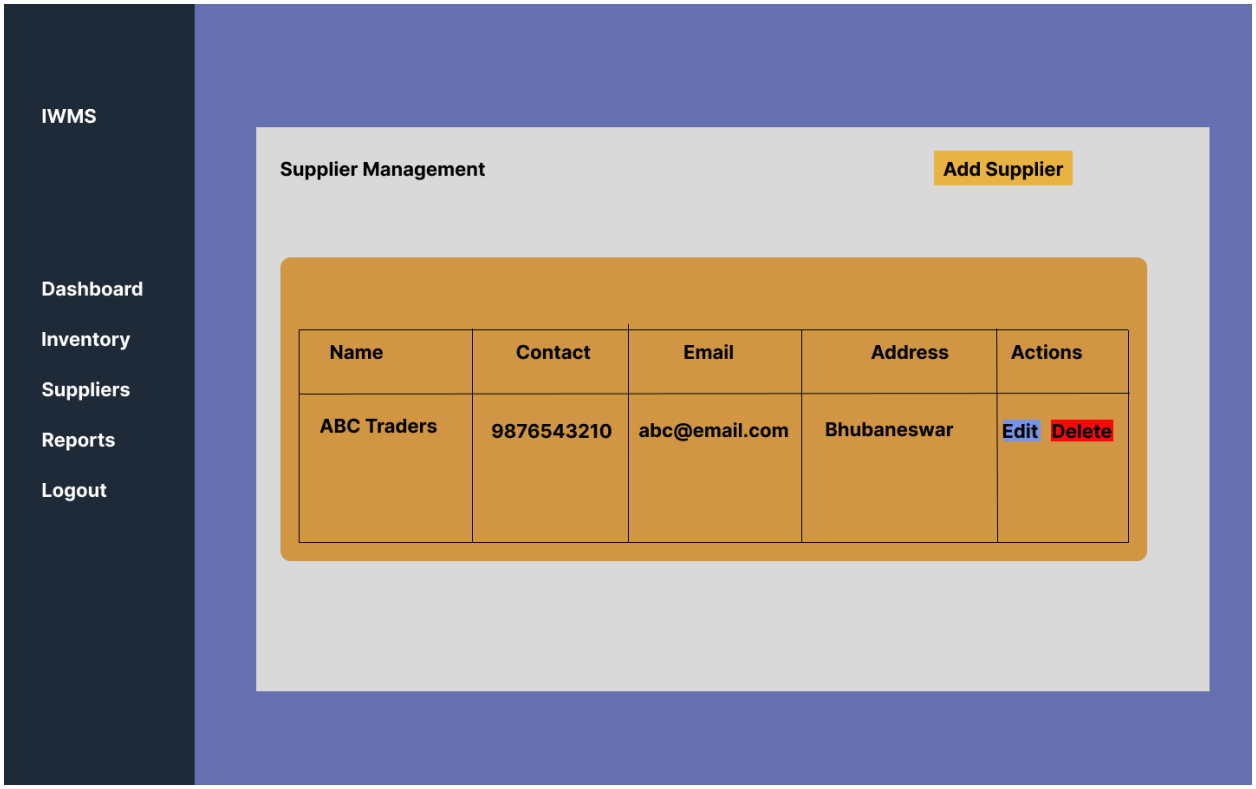


Figure 3-6: Supplier Management Screen

Figure 3-7: Low Stock Alert Screen

3.3 Minimal UI Notes

The user interface follows a clean dashboard-oriented layout with consistent navigation across screens. Action buttons are highlighted using primary colors, and tables are structured for readability and clarity.

The system is designed with:

- Role-based access control
- Minimalist design principles
- Responsive layout
- Clear visual hierarchy